

Isaac Mubiru

isaacmubiru99@gmail.com | +256702360601 | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 |
Kampala, Uganda (UTC+3)

PROFESSIONAL SUMMARY

Backend engineer with 6 years building TypeScript/Node.js serverless functions and real-time data systems on GCP, AWS, and Firebase — including 4+ years shipping daily on isolate-based JavaScript function runtimes (Firebase Cloud Functions). Led the zero-downtime migration of 20+ production applications to Firebase/GCP, reporting to the CTO, and built a real-time two-way MongoDB↔Firestore sync pipeline on Google Pub/Sub that carried parallel production traffic. Experienced operating Kubernetes and Kafka-based microservices, WebSockets/streaming workloads, and scheduled background jobs at multi-million-record scale. Intermediate Rust with a systems background (BSc Computer Engineering) and a strong interest in JavaScript runtime internals (Deno, V8 isolates) — now moving from power user of edge function runtimes to building them.

CORE COMPETENCIES

TypeScript & Node.js

Serverless Functions (Cloud Functions Gen2)

Real-time Data Sync (Pub/Sub)

Kubernetes & Docker

GCP & AWS

Event Streaming (Kafka, WebSockets)

Zero-Downtime Migrations

Rust (intermediate)

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote (from UTC+3)

- Implemented **real-time two-way sync** between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing, carrying parallel production traffic with no data loss during a live migration
- Led the **end-to-end migration of 20+ applications** from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO
- Saved the company **\$5,000/month** by migrating all services off AWS; moved file storage from Amazon S3 to GCS with Python tooling
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion
- Implemented Express Shipping, generating an additional \$40 revenue per order

MTailor

Mar 2022 – Jul 2022

Contractor (Backend Migration) — US Remote

- Built the migration pipeline reading from MongoDB and writing to Firestore after complex processing logic
- Wrote SDK documentation (Node.js and Python) for fellow engineers and trained the Ops team on the new Firebase stack

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Extended backend APIs using **Firebase Cloud Functions** and Express.js for a real-time stock-price PWA
- Built scheduled background jobs querying **2M+ Firestore records** to keep customer prices current from Morningstar APIs
- Shipped responsive UI pages with React and Tailwind

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for DeFi applications using Web3 and Node.js; delivered React UIs from Figma under tight timelines

CodeBits

Jan 2020 – Jul 2021

Software Engineer (led team of 4) — Uganda

- Architected a **microservices backend** for FIDA Uganda's case management system using Apache Kafka, Docker, and Kubernetes
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC
- Developed a cross-platform mobile app (React Native/Expo) with Firebase push notifications

PROJECTS

Real-time Pub/Sub Sync Pipeline

Live MongoDB↔Firestore two-way sync supporting parallel production traffic throughout a months-long migration — distributing data safely to where it is read.

Node.js, Google Pub/Sub, Firestore, MongoDB

Healthcare WhatsApp Chatbot

Webhook-driven chatbot integrating OpenAI and Google Vision for prescription OCR via the WhatsApp Business API — webhook receivers and AI inference in serverless functions.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Marketplace Recommendation Engine

Collaborative and content-based filtering engine in MongoDB for service provider matching.

Node.js, MongoDB

EDUCATION

BSc Computer Engineering — [Makerere University](#)

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, Firebase, GCP, Python, MongoDB, SQL, React, Linux, Git **Intermediate:** Rust, Go, AWS, React Native

Infrastructure: Kubernetes, Docker, Apache Kafka, Google Pub/Sub, gRPC **Languages:** English, Kiswahili